HUJI Research Monitor

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EXECUTIVE SUMMARY

This weekly digest highlights groundbreaking advancements in neurodegenerative disease management and novel cancer immunotherapies from Hebrew University researchers. One standout project features a second-generation AI system designed for personalized Parkinson's disease management, showing promising results in clinical trials. Simultaneously, a significant discovery reveals how a specific bacterial protein can activate natural killer cells to target and destroy head and neck cancer cells. These innovations offer substantial commercial potential for investors and industry partners looking to drive the next wave of therapeutic and diagnostic solutions.

#1

A feasibility open-label clinical trial utilizing second-generation artificial intelligence based on the constrained-disorder principle in patients with Parkinson's disease.

31/50

Main Researcher: Ilan Y

[Software/AI] [Neuroscience] [Clinical] [Medical Device]

"Novel AI platform poised to revolutionize personalized Parkinson's disease management and integrate with existing medical devices."

WHY NOW

The rapidly growing market for digital health solutions and neurodegenerative disease management offers a timely entry point. This advanced AI can enhance current diagnostic and therapeutic strategies, providing a clear path to market through licensing or partnerships.

COMMERCIAL ANGLE

Commercialization could focus on licensing the AI algorithms for personalized Parkinson's disease management, potentially integrating with existing wearable sensors or developing a companion diagnostic to identify suitable patients.

https://pubmed.ncbi.nlm.nih.gov/41853628/

#2

RadD from <i>Fusobacterium nucleatum</i> engages NKp46 to promote antitumor cytotoxicity.

27/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical] [Genomics]

"Breakthrough immunotherapy leveraging bacterial protein to activate NK cells for targeted head and neck cancer treatment."

WHY NOW

The robust and expanding field of cancer immunotherapy provides a fertile ground for this novel mechanism. With a clear target indication, this discovery has immediate potential for drug development partnerships to address an unmet need.

COMMERCIAL ANGLE

Developing therapies that enhance NKp46-RadD interaction, or engineer *F. nucleatum* strains with optimized RadD expression, could offer a novel immunotherapeutic approach for specific cancer types like head and neck cancer.

https://europepmc.org/article/MED/42065370